Sabaa at DialectSentEval Shared Task Subtask-2: Classifier-Guided Candidate Reranking for Arabic Dialectal Sentiment Swap

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Abstract

Sentiment swap asks a system to rewrite a sentence so that its polarity is inverted while everything else stays the same, two halves of an instruction that pull against each other. This paper describes how the Sabaa system navigates that trade-off for Arabic dialects in Subtask 2 of DialectSentEval 2026. Rather than asking one model for a single correct rewrite, we ask two fine-tuned Arabic sequence-to-sequence models to each propose several, then choose among the pooled proposals with a scoring function built from the signals the task grades us on. The system scores 49.07 BLEU, 69.33 chrF, and 0.7447 sentiment accuracy on the official test set. That last figure is the most informative result here: the system places 90.37% of its outputs at the polarity it was aiming for, and the shortfall traces to inferring each sentence's original polarity, which the test data withholds. Measured against gold labels on development, that inference is correct only 74.07% of the time, and sixteen points of sentiment accuracy separate our submission from a perfect estimator. [1]

1 Introduction

Consider the Egyptian Arabic sentence كل حاجه جميله ("everything is beautiful"). Asked to invert its sentiment, a system could return كل حاجه ضلمه ("everything is bleak"), changing one adjective, or a fluent but unrelated complaint. Both flip the polarity; only the first preserves the content, phrasing, and dialect. Subtask 2 of DialectSentEval 2026 (Ezzini et al., 2026a) scores both properties at once, sentiment accuracy through a fixed classifier and content preservation through BLEU and chrF, rewarding a system precisely for preferring the first output over the second.

[1] https://github.com/mohamedkamalsabaa/Sabaa_DialectSentEval_2026_Subtask-2

This framing shapes our design. A model trained only to maximise likelihood has no reason to prefer a minimal edit over a wholesale rewrite, since both are fluent continuations of the same instruction. What it does reliably produce is a *set* of plausible rewrites, some minimal and some not. Our system therefore separates generation from selection: two fine-tuned Arabic encoder–decoder models propose, and a scoring function assembled from the task's own evaluation signals decides. Pooling candidates from two models pretrained on different corpora widens that set, and their proposals tend to fail in different ways, so a candidate one model truncates the other often does not.

The system obtains 0.9564 sentiment accuracy, 52.26 BLEU, and 70.64 chrF on the development leaderboard, and 49.07 BLEU, 69.33 chrF, and 0.7447 sentiment accuracy on test. The drop in sentiment demands an explanation, and supplying one leads to the paper's main finding (§5.3): the generator is not what changed. The test set withholds the source polarity, forcing the system to infer the swap direction, and an incorrect direction guarantees an incorrect output however good the rewrite that follows.

2 Background

What the task asks for. Each instance pairs a source sentence with its polarity, and the system must return a sentence of the opposite polarity that otherwise preserves the original. The following pair from the training data shows the shape of the intended output:

Source (Positive):
بنقول لنفسنا كل حاجه جميله
Target (Negative):
بنقول لنفسنا كل حاجه ضلمه

One word carries the sentiment and one word changes; the dialectal spelling, the subject, and the

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sentence frame all survive. This is the behaviour our reranker is built to select for.

Data. We train on MA'AKS (Mughaus et al., 2026), a manually-curated parallel corpus for Arabic sentiment swap, which supplies 6,752 training and 964 development pairs carrying `id`, `source`, `source_polarity`, and `target`; training polarities are mildly imbalanced at 3,731 negative to 3,021 positive.

The 1,931-sentence test set differs in a way that matters a great deal: it supplies only `id` and `source`. The target is withheld, as expected, but so is `source_polarity`, so a system must work out the swap direction for itself (§3.4). The predecessor AraSentEval task (Ezzini et al., 2026b) established this formulation and we follow its evaluation protocol.

3 System Overview

3.1 Doubling the training data for free

Sentiment swap is symmetric: if t is a valid polarity-inverted rewrite of s, then s is equally a valid rewrite of t. Every training instance therefore contains a second that costs nothing to extract, so for each pair (s, t) with source polarity p we add (t, s) with polarity $\overline{p}$ under an offset identifier. This doubles the data to 13,504 pairs and balances the polarity distribution exactly at 6,752 each, so the model sees both swap directions equally often, which matters because at test time it performs both.

3.2 Two generators, one instruction

We fine-tune two Arabic-specific encoder–decoder models on this augmented data: **AraT5v2-base-1024** (Nagoudi et al., 2022), a 220M-parameter Arabic T5 pretrained on both Modern Standard Arabic and dialectal text, and **AraBART** (Kamal Eddine et al., 2022), a 139M-parameter Arabic BART pretrained on news. Both receive an identical protocol and instruction, so the only difference between the two candidate streams is the model behind it:

> اعكس قطبية النص من `{src_pol}` إلى `{tgt_pol}`: `{source}`
>
> ("Invert the polarity of the text from `{src_pol}` to `{tgt_pol}`")

The instruction is Arabic by design: both models were pretrained exclusively on Arabic, and an English frame would ask them to parse a token sequence unlike anything in their pretraining before reaching the sentence itself.

3.3 Choosing among proposals

At inference each model produces k candidates by beam search, and the two sets are pooled into a single candidate set $\mathcal{C}$ of size $2k$. Every candidate in this pool is then scored on three properties.

The first is whether it carries the right sentiment. We ask the official evaluation classifier, CAMeLBERT-DA (Inoue et al., 2021), writing $\hat{y}(c)$ for its label, $\sigma(c)$ for its confidence, and τ for the target polarity: $S_{\text{sent}}(c) = +\sigma(c)$ if $\hat{y}(c) = \tau$ and $-\sigma(c)$ otherwise. Signing the confidence rather than using a binary indicator penalises a confidently wrong candidate more heavily than an uncertain one.

The second property is whether the candidate resembles its peers. Following the Minimum Bayes Risk principle (Eikema and Aziz, 2020, 2022), $S_{\text{cons}}(c)$ is the mean chrF between c and every other member of $\mathcal{C}$, scaled to $[0, 1]$. The justification is empirical: pool outliers were overwhelmingly truncations, repetitions, and hallucinations rather than unusually good rewrites.

The third property is whether the content survived, measured as the chrF of the candidate against the source, $S_{\text{cont}}(c) = \text{chrF}(c, s)/100$. This is what makes minimal edits attractive to the selector, since a rewrite changing one word scores far higher against the source than one that rephrases.

The output is the candidate maximising $w_1 S_{\text{sent}} + w_2 S_{\text{cons}} + w_3 S_{\text{cont}}$. A candidate decoding to an empty string is replaced by the source, guaranteeing every input a non-empty, on-topic output.

3.4 Recovering the swap direction

Everything above assumes τ is known. On development it is, since `source_polarity` is supplied. On test it is not, so we recover it with the same classifier we use for reranking. CAMeLBERT-DA is three-way, and its Neutral class names no direction to swap in, so we discard the Neutral logit and read the source polarity off the remaining two: assigning $\hat{p}(s) =$ Positive when $P_{\text{pos}}(s) \geq P_{\text{neg}}(s)$ and Negative otherwise, then taking $\tau = \overline{\hat{p}(s)}$. On test this assigns 977 sentences a positive source polarity and 954 a negative one. This step is the only substantive difference between our development and test pipelines, and §5.3 shows it is where most of our test-set sentiment loss originates.

4 Experimental Setup

Both generators are trained on the full augmented training split of 13,504 pairs under identical settings: maximum sequence length 128, per-device batch size 4 with two gradient accumulation steps, AdamW at learning rate 5×10^{-5}, 500 warmup steps, weight decay 0.01, 15 epochs, and mixed-precision training. Label smoothing is disabled, for reasons taken up in §5.2. Decoding uses `no_repeat_ngram_size` 3, repetition penalty 1.2, and length penalty 1.0.

Two settings differ between the development and test submissions. Development used $k = 5$ per model with weights $(0.6, 0.2, 0.2)$; the test run widened the pool to $k = 8$ and rebalanced to $(0.45, 0.10, 0.45)$, giving sentiment and content near-equal influence since the task scores both. These changes were made together, and the test set also introduced polarity inference, so we cannot attribute the difference between the runs to any one of them.

We use Transformers 4.44.0 (Wolf et al., 2020) and sacreBLEU (Post, 2018) for BLEU (Papineni et al., 2002) and chrF (Popović, 2015), on 2×NVIDIA T4 GPUs, where the complete pipeline takes about 2.5 hours.

One caveat applies to every content figure we compute ourselves: our offline reimplementation of BLEU and chrF did not reproduce the official scorer under any pairing or aggregation tried, including output-against-reference and output-against-input at sentence and corpus level. We therefore report official leaderboard numbers as headline results and label our own measurements as internal diagnostics.

5 Results

5.1 Official scores

Table 1 reports every official score we obtained. Content preservation is stable between the two splits, losing 3.2 BLEU and 1.3 chrF on the larger, dialect-oriented test set. Sentiment accuracy behaves very differently, falling by more than twenty points, and that asymmetry is what §5.3 examines.

5.2 What each component contributes

Table 1 follows the system as it was built. Replacing Adafactor with AdamW and training for 15 rather than 10 epochs is the single largest change, adding 5.9 points of sentiment accuracy and 1.1 BLEU, entirely a matter of optimisation rather than architecture.

Configuration	Sent.	BLEU	chrF
Development split			
Single generator, Adafactor	0.8651	51.17	70.11
+ AdamW, 15 epochs	0.9243	52.30	71.08
- smoothing, + consensus	0.9284	**52.61**	**71.08**
+ second generator	**0.9564**	52.26	70.64
Test split, submitted system			
inferred polarity, k=8	0.7447	49.07	69.33

Table 1: Official leaderboard scores. **Sent.** is sentiment style accuracy in $[0, 1]$; BLEU and chrF are on a 0–100 scale. Development rows are cumulative, each adding to the one above. The final row is the submitted test run, which also widened the pool to k=8 and reweighted to $(0.45, 0.10, 0.45)$. One run per configuration.

Removing label smoothing and adding the consensus term together produce the best content scores we recorded. The label smoothing result initially surprised us, since smoothing lowered training loss as usual. The explanation is that our system consumes not the single best output but a ranking over many: label smoothing flattens the output distribution (Meister et al., 2020), and a flatter distribution discriminates less sharply among candidates, so a regulariser that helps single-best decoding can hurt reranked decoding.

Adding the second generator buys 2.8 points of sentiment accuracy for 0.35 BLEU and 0.44 chrF. We judged this favourable given that the task scores both axes, and submitted it, while flagging that it was a judgement call: without a validated local copy of the official metric we could not tune the trade-off empirically.

5.3 Where the sentiment score went

Sentiment accuracy falls from 0.9564 on development to 0.7447 on test. The generators, the training data, and the reranking procedure are the same in both runs, so an explanation is required.

Two measurements localise it. The first comes from instrumentation in the test pipeline: we recorded how often the output carried the polarity the system was *aiming* for, the opposite of the polarity it had inferred. That figure is 1,745 of 1,931, or $h = 0.9037$. The system hit its own target on test about as reliably as on development; what it could not guarantee was that its own target was right.

The second measurement addresses that. The development split supplies gold `source_polarity`,

so the inference step of §3.4 can be run there and scored directly. It is correct for 714 of 964 sentences, an accuracy of 0.7407, and the errors are near-symmetric rather than a bias toward one class: 140 Negative sources are read as Positive against 110 in the reverse direction, giving recall of 0.7373 and 0.7448 respectively. Misclassified sources are somewhat shorter than average, at 10.7 words against 12.5.

Combining the two measurements tests the decomposition rather than assuming it. Let p be the accuracy of polarity inference. A right direction scores correct with probability h; a wrong one, if the system reached its own mistaken target, would score correct only in the residual $1 - h$, giving $p\,h + (1-p)(1-h) = 0.6943$ against an official 0.7447.

The gap is informative. Writing r for the true probability of scoring correct on a wrong-direction item, solving $0.7447 = p\,h + (1-p)\,r$ at $p = 0.7407$ gives $r = 0.29$, three times what the simple account predicts: something recovers roughly a third of the misdirected items. A plausible mechanism, which we flag as a hypothesis consistent with the arithmetic rather than an established fact, is that both generators saw nothing but polarity-flipping pairs and so tend to flip regardless of the instruction, which on a misdirected item is correct.

The ceiling does not depend on any of this: under every value of p considered, a perfect polarity estimator would have left the official score at $h = 0.9037$, so sixteen points separate our submission from that ceiling and none are attributable to the generators. One caveat is that p is measured on development, and the test split may be easier or harder to classify.

5.4 Error analysis

Our internal content diagnostic rises monotonically with source length, from 58.6 chrF on inputs of four words or fewer ($n = 143$) to 80.7 on inputs of 26 or more ($n = 102$): in a short sentence the word that must change is a large fraction of the text, so even a correct minimal edit registers as a large proportional change. The lowest-scoring test item shows this: جو مشاوي [GRINNING] becomes جو حلو [GRINNING], a defensible rewrite scoring 11.3 chrF.

Three failure modes recur. In 39 cases (2.0%) the output is byte-identical to the input, no swap having been performed; these cluster in sentences whose sentiment is idiomatic rather than lexical, leaving no obvious substitution target. On some short inputs the model over-edits, replacing the sentence rather than adjusting it, as in وين رحتي [PENSIVE] becoming فرحانتي . Finally, it sometimes flips the textual sentiment while leaving a contradictory emoji in place. Replacing each emoji with a textual gloss before training looks the more promising remedy, since it keeps the signal inside the token stream; stripping emoji discards it, and flipping them post hoc needs a polarity lexicon the task does not supply.

Set against these, the successful outputs are precisely the minimal edits the reranker was designed to favour: كل حاجه ضلمه becomes كل حاجه جميله at 85.5 chrF, changing one adjective. Across the test set the mean output/input length ratio is 0.993, direct evidence that the selection stage works, choosing minimal edits from a pool containing both those and wholesale rewrites.

6 Conclusion

We have described a system that treats sentiment swap as a selection problem as much as a generation problem: two Arabic sequence-to-sequence models propose a pool of rewrites, and a scoring function built from the task's own evaluation signals chooses among them. It scores 49.07 BLEU, 69.33 chrF, and 0.7447 sentiment accuracy on the official test set, with the shortfall lying in recovering the swap direction rather than in the rewriting itself. That points to the clearest next step: rather than committing to a direction before generation, a system could generate in both and let the reranker choose, turning an unrecoverable error into an ordinary ranking decision.

Limitations. Each configuration was run once, so seed variance is not separable from real effects, and the last three development rows of Table 1 differ by little enough for this to matter. The reranking weights were judged rather than tuned, since no validated local copy of the official metric was available. The development and test runs differ in pool size, weighting, and polarity inference at once, preventing clean attribution.

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